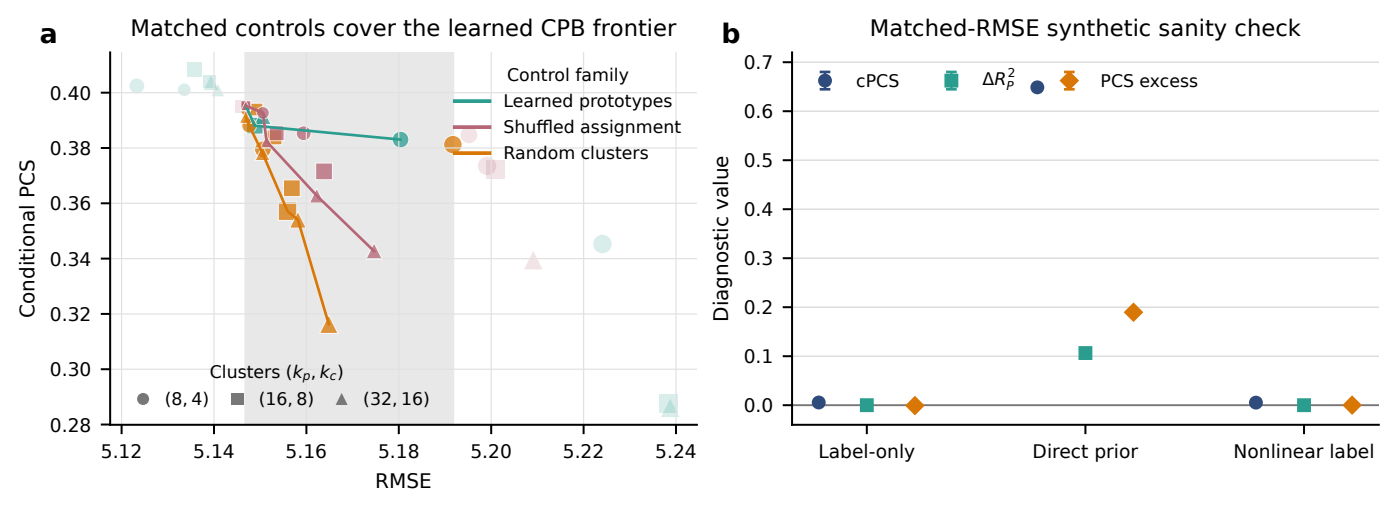

a
Matched controls cover the learned CPB frontier
Control family
Learned prototypes
Shuffled assignment
Random clusters
Clusters (k_p, k_c)
(8, 4)
(16, 8)
(32, 16)
Conditional PCS
RMSE
0.40
0.38
0.36
0.34
0.32
0.30
0.28
5.12
5.14
5.16
5.18
5.20
5.22
5.24
b
Matched-RMSE synthetic sanity check
cPCS
ΔR_p^2
PCS excess
Diagnostic value
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Label-only
Direct prior
Nonlinear label